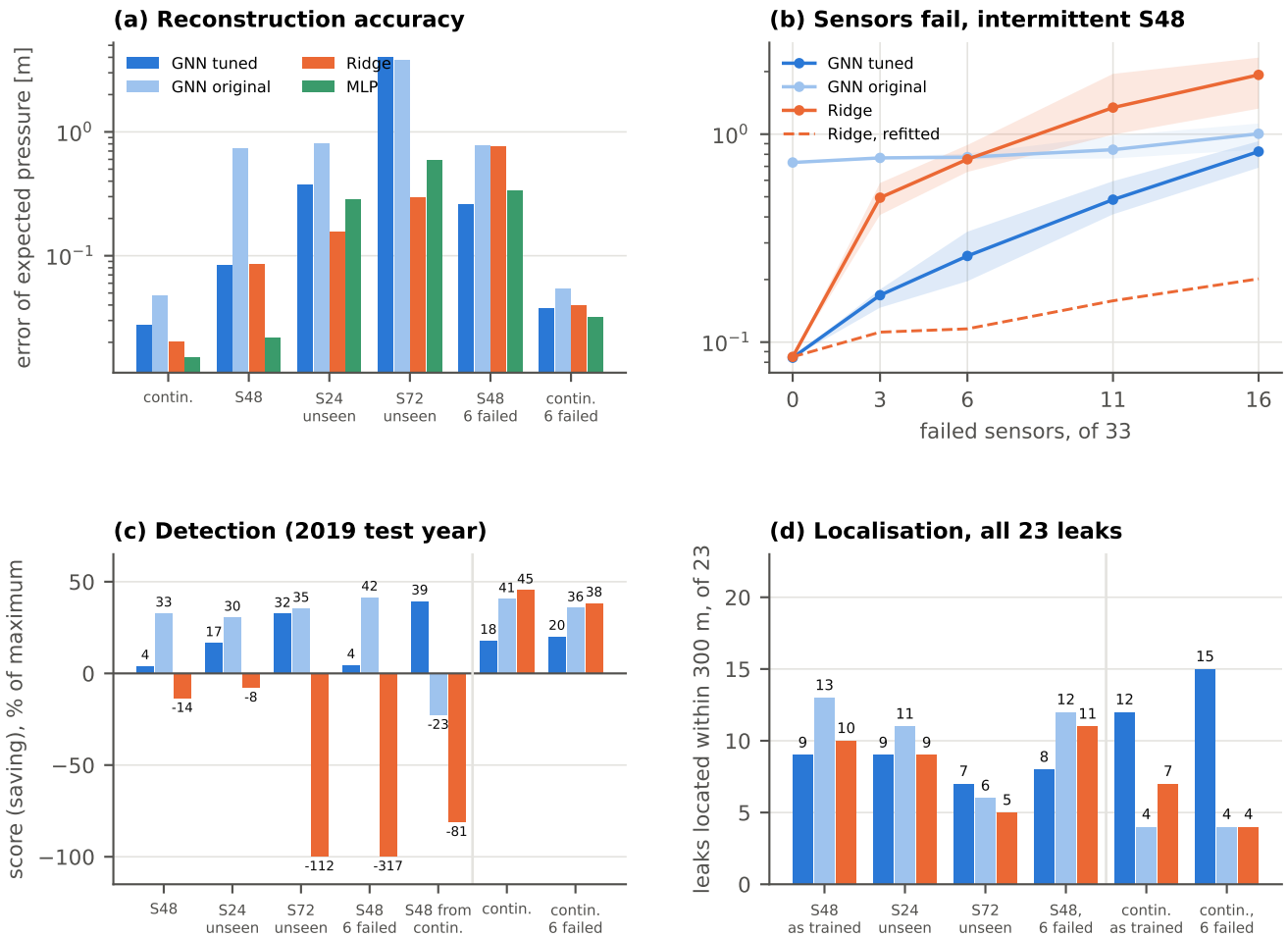
(a) Reconstruction accuracy
GNN tuned
GNN original
Ridge
MLP
error of expected pressure [m]
10^0
10^{-1}
contin.
S48
S24 unseen
S72 unseen
S48 6 failed
contin. 6 failed
(b) Sensors fail, intermittent S48
GNN tuned
GNN original
Ridge
Ridge, refitted
10^0
10^{-1}
0
3
6
11
16
failed sensors, of 33
(c) Detection (2019 test year)
score (saving), % of maximum
50
0
−50
−100
4
33
-14
17
30
-8
32
35
-112
4
42
-317
39
-23
-81
18
41
45
20
36
38
S48
S24 unseen
S72 unseen
S48 6 failed
S48 from contin.
contin.
contin. 6 failed
(d) Localisation, all 23 leaks
leaks located within 300 m, of 23
20
15
10
5
0
9
13
10
9
11
9
7
6
5
8
12
11
12
4
7
15
4
4
S48 as trained
S24 unseen
S72 unseen
S48, 6 failed
contin. as trained
contin., 6 failed